Figure 5.5. The bias-variance tradeoff. Increasing model complexity can reduce bias, but it will increase variance. Reducing variance will increase bias.

Figure 5.6. The general overview of actor-critic models. First, the actor predicts the best action and chooses the action to take, which generates a new state. The critic network computes the value of the old state and the new state. The relative value of St+1 is called its advantage, and this is the signal used to reinforce the action that was taken by the actor.

Figure 5.7. A generic recurrent neural network (RNN) layer processes a sequence of data by incorporating its previous output with the new input. The input on the left, along with a previous output is fed into an RNN module, which then produces an output. The output is fed back into the RNN on the next time step, and a copy may be fed into another layer. An RNN will not work properly with single experiences in an experience replay buffer since it needs to work on sequences of experiences.

Figure 5.8. The most common form of training a deep learning model is to feed a batch of data together into the model to return a batch of predictions. Then we compute the loss for each prediction and average or sum all the losses before backpropagating and updating the model parameters. This averages out the variability present across all the experiences. Alternatively, we can run multiple models with each taking a single experience and making a single prediction, backpropagate through each model to get the gradients, and then sum or average the gradients before making any parameter updates.

Figure 5.9. A simple multiprocessing example. We want to more efficiently square all the numbers in an array. Rather than squaring each element one by one, we can split the array into two pieces and send each piece to a different processor that will square them simultaneously. Then we can recombine the pieces back into a single array.

Figure 5.10. An actor-critic model produces a state value and action probabilities, which are used to compute an advantage value and this is the quantity that is used to train the model rather than raw rewards as with just Q-learning.